

ZAYAN AHMAD GHOUS

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PROFESSIONAL SUMMARY

BSCS student (4th semester) with hands-on experience in Machine Learning, Data Engineering, and MLOps. Skilled in building data preprocessing pipelines, deploying machine learning models, and developing REST APIs using Python, TensorFlow, and FastAPI. Experienced with cloud-based ML applications and focused on delivering scalable, efficient solutions. Seeking an opportunity to apply technical expertise while gaining industry experience.

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C
- Machine Learning & AI:** TensorFlow, Keras, scikit-learn, NumPy, Pandas, Convolutional Neural Networks (CNN), classification, regression, model optimization, cross-validation
- Data Engineering:** Data preprocessing, feature engineering, exploratory data analysis (EDA), data cleaning, feature scaling, categorical encoding, data augmentation
- API Development:** FastAPI, Django REST Framework, REST API design, HTTP methods, API testing (Postman), Gradio
- Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, Hugging Face Spaces, Streamlit, Linux terminal
- Data Processing Tools:** Pandas (data manipulation), NumPy (numerical operations), Matplotlib (data visualization)
- Currently Learning:** MLOps pipelines, Kubernetes, LLM integration, advanced API frameworks

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern University of Engineering & Technology (UET) Lahore

May 2025 – July 2025

- Developed and trained multiple machine learning models including Decision Trees, Random Forests, and Neural Networks using Python and scikit-learn, achieving 85%+ accuracy on real-world datasets
- Built end-to-end data preprocessing pipelines including missing value handling, feature scaling, categorical encoding, and data transformation for datasets with 10,000+ samples
- Worked on data engineering tasks including data cleaning, ETL workflows, dataset integration, and automated preprocessing pipelines using Pandas and NumPy
- Designed CNN-based image classification models for MRI brain scan analysis using TensorFlow/Keras with data augmentation techniques to improve model robustness
- Deployed machine learning applications using FastAPI and Gradio for real-time inference and interactive web-based predictions
- Performed exploratory data analysis (EDA), feature engineering, and model evaluation while managing projects with Git/GitHub and Jupyter Notebook

PROJECTS

Brain Tumor Detection (CNN + FastAPI + Deployment)

Live: huggingface.co/spaces/Ghous/brain_tumor

- Built a CNN-based brain tumor classification model achieving 88% accuracy on MRI dataset (500+ images) using TensorFlow/Keras
- Improved model generalization using data augmentation, dropout regularization, batch normalization, and early stopping
- Developed REST API using FastAPI for real-time inference and integrated with a Gradio UI for user-friendly predictions
- Deployed end-to-end ML application on Hugging Face Spaces for public web access
- Tech Stack: Python, TensorFlow, Keras, FastAPI, Gradio, scikit-learn

House Price Prediction API (End-to-End MLOps Pipeline)

Image Link: [zayanahmad123/house-price-api](https://zayanahmad123.github.io/house-price-api/)

- Built a house price prediction model using scikit-learn with feature engineering, outlier removal, and categorical encoding on a real-world housing dataset
- Developed automated preprocessing pipelines using Pandas, SciPy, and MEstimateEncoder for reliable model training and inference
- Created a FastAPI-based REST API for real-time house price predictions with production-ready inference workflows
- Containerized the complete application using Docker and published the image on Docker Hub for reproducible deployment
- Tech Stack: Python, scikit-learn, Pandas, NumPy, FastAPI, Docker, SciPy

EDUCATION

Board of Intermediate and Secondary Education

Intermediate in Computer Sciences (Punjab Group of colleges)

Minhaj University Lahore

BS in Computer Sciences

Lahore, Pakistan

2022 – 2024

Lahore, Pakistan

2024 – Present